

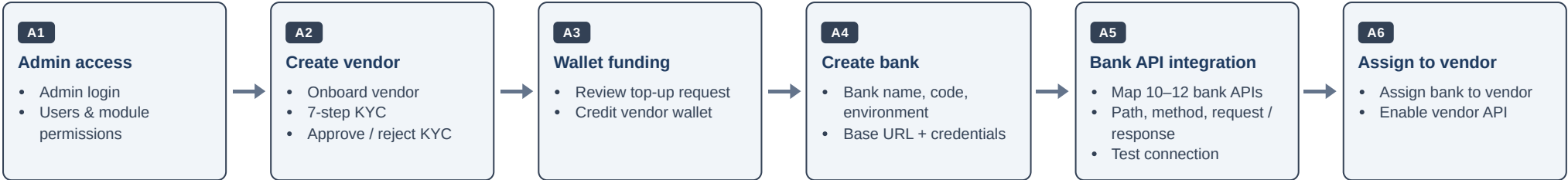
FinPay – India → Nepal Remittance & Vendor Payment Platform

Bank-facing system architecture and end-to-end process flow

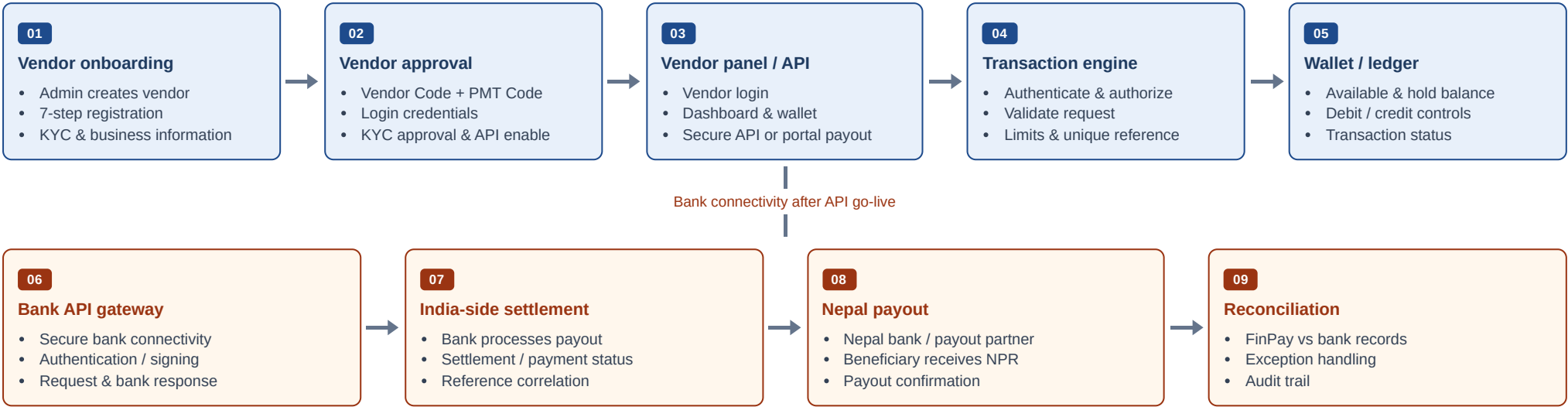
Page 1 of 4

☐ Admin operations ☐ Vendor / FinPay engine ☐ Bank settlement / Nepal payout

1. Admin operations



2. Vendor payout flow



Admin creates vendors, funds wallets, creates banks, integrates 10–12 bank APIs and assigns them to vendors. Vendors never receive bank credentials. India-side settlement and Nepal NPR credit follow after bank API certification.

MODULE	PURPOSE / KEY FUNCTIONS	PRIMARY CONTROL
Admin operations	Admin login; staff users and module permissions; create vendor; KYC approve/reject; wallet top-up credit; create bank; map 10–12 bank APIs; assign bank to vendor and enable API.	Role-based modules; encrypted bank secrets; audit trail
Vendor onboarding	7-step vendor registration; business, legal, promoter/director, team/IT, business-plan/evaluation; resume from pending step; admin KYC approve/reject.	Role-based access; validation; audit trail
Vendor authentication	Vendor login; vendor-specific dashboard. API uses HMAC-SHA256 (secret never sent in the request) plus IP allowlist.	Password hashing; authentication; authorization
Vendor wallet	Available and hold balance; admin-approved top-up; transaction-limit and balance checks before processing.	Controlled debit/credit; transaction integrity
Transaction management	Create transaction, unique reference, amount, beneficiary, service (IMPS/NEFT/RTGS), type and status lifecycle.	Validation; reference control; status tracking
API layer	Vendor-to-FinPay APIs (balance, banks, payouts, /api/v1/bank/{code}/{slug}). Bank APIs are mapped by admin and are not given raw to the vendor.	TLS; HMAC; IP allowlist; rate limiting; secret isolation
Bank integration	Approved transaction sent to mapped bank API; response/webhook/status handled by FinPay. Admin stores encrypted bank credentials and assigns the bank to vendors.	Secure connectivity; request/response validation
Nepal payout	Bank / payout partner completes beneficiary-side NPR payout after India-side settlement.	Partner controls; status confirmation
Reconciliation	Match FinPay transactions with bank/settlement records and resolve exceptions.	Reference matching; audit logs
Reports & monitoring	Vendor reports, transaction reports, settlement reports, operational alerts and audit activity.	Access control; monitoring; event history

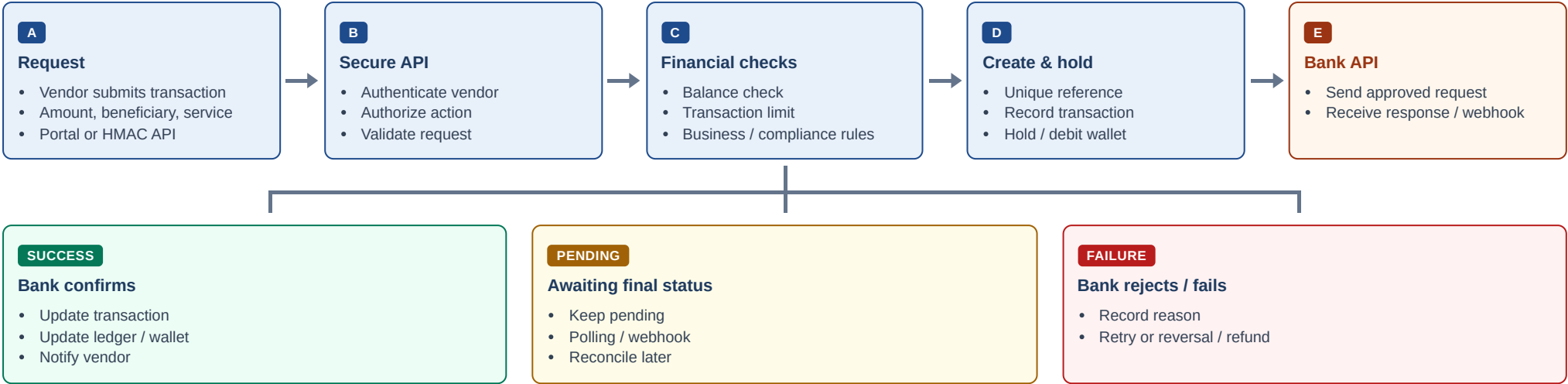
LAYER	TECHNOLOGY / APPROACH	PURPOSE
Application	PHP 8.2, Laravel 12, Livewire 3 — Admin & Vendor interfaces	Business logic, onboarding, dashboards and workflow
Database	MySQL relational database	Vendor, wallet, transaction, registration and audit-related data
Cache	Database cache	HMAC nonce replay protection and rate limits
API	REST JSON; HTTPS/TLS; HMAC-SHA256 vendor signing	Vendor integration and bank outbound calls
UI	Blade + Tailwind CSS	Admin and vendor consoles
Authentication	Separate admin/vendor guards; hashed passwords; encrypted sessions	Verify identity and enforce access boundaries
Secrets	Bank credentials encrypted at rest; not in source code; never shown to vendors	Reduce credential exposure

Security control framework

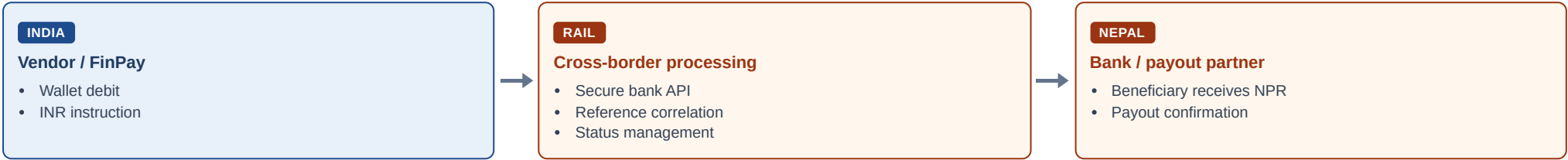
SECURITY AREA	CONTROL / DESIGN PRINCIPLE
Transport security	HTTPS/TLS for vendor API, bank outbound calls and webhooks. HSTS when the application is served over TLS.
Authentication	Portal session login (regenerate on login). API: HMAC-SHA256 over timestamp, nonce, method, path and body hash. Secret is never transmitted in the request.
Authorization	Admin module permissions; vendor-scoped data; encrypted IDs in URLs.
API protection	Request validation, mandatory IP allowlist, rate limiting, generic auth errors, body-size cap, redacted API logs.
Transaction integrity	Unique transaction/reference identifiers, wallet lock, status lifecycle, pending / success / failed handling.
Data protection	Password hashing; least-privilege access; encrypted bank secrets; account numbers redacted in logs.
Outbound safety	Bank and webhook URLs restricted (HTTPS in production; no private/metadata hosts; no open redirects).
Audit & monitoring	Administrative, API and transaction events retained for investigation and reconciliation.
Operational resilience	Pending states, failure handling, polling/webhook and reconciliation for external bank responses.

Important: Controls describe the implemented architecture. Regulatory compliance, certification, VAPT and bank-specific production requirements should be validated before go-live.

1. Request processing



2. India → Nepal corridor



Document purpose: Architecture and process-flow for banking / integration stakeholders. API specifications, settlement models, limits, message formats, certificates and SLAs are finalized with the bank during integration.